

Psychiatry

“Words of comfort, skillfully administered, are the oldest therapy known to man.”

—Louis Nizer

“All men should strive to learn before they die what they are running from, and to, and why.”

—James Thurber

“Man wishes to be happy even when he so lives as to make happiness impossible.”

—St. Augustine

“It is far more important to know what sort of person has a disease, than to know what sort of disease a person has.”

—Hippocrates

► Psychology	524
► Pathology	526
► Pharmacology	542

► PSYCHIATRY—PSYCHOLOGY

Classical conditioning	Learning in which a natural response (salivation) is elicited by a conditioned, or learned, stimulus (bell) that previously was presented in conjunction with an unconditioned stimulus (food).	Usually deals with involuntary responses. Pavlov's classical experiments with dogs—ringing the bell provoked salivation.									
Operant conditioning	Learning in which a particular action is elicited because it produces a punishment or reward. Usually deals with voluntary responses.										
Reinforcement	Target behavior (response) is followed by desired reward (positive reinforcement) or removal of aversive stimulus (negative reinforcement).										
Extinction	Discontinuation of reinforcement (positive or negative) eventually eliminates behavior. Can occur in operant or classical conditioning.										
Punishment	Repeated application of aversive stimulus (positive punishment) or removal of desired reward (negative punishment) to extinguish unwanted behavior (Skinner's operant conditioning quadrant).	<table> <tr> <th></th><th>Increase behavior</th><th>Decrease behavior</th></tr> <tr> <th>Add a stimulus</th><td>Positive reinforcement</td><td>Positive punishment</td></tr> <tr> <th>Remove a stimulus</th><td>Negative reinforcement</td><td>Negative punishment</td></tr> </table>		Increase behavior	Decrease behavior	Add a stimulus	Positive reinforcement	Positive punishment	Remove a stimulus	Negative reinforcement	Negative punishment
	Increase behavior	Decrease behavior									
Add a stimulus	Positive reinforcement	Positive punishment									
Remove a stimulus	Negative reinforcement	Negative punishment									

Transference and countertransference

Transference	Patient projects feelings about formative or other important persons onto physician (eg, psychiatrist is seen as parent).
Countertransference	Doctor projects feelings about formative or other important persons onto patient (eg, patient reminds physician of younger sibling).

Ego defenses Mental processes (unconscious or conscious) used to resolve conflict and prevent undesirable feelings (eg, anxiety, depression).

IMMATURE DEFENSES	DESCRIPTION	EXAMPLE
Acting out	Expressing unacceptable feelings and thoughts through actions.	Tantrums.
Denial	Avoiding the awareness of some painful reality.	A patient with cancer plans a full-time work schedule despite being warned of significant fatigue during chemotherapy.
Displacement	Redirection of emotions or impulses to a neutral person or object (vs projection).	A teacher is yelled at by the principal. Instead of confronting the principal directly, the teacher goes home and criticizes her husband's dinner selection.
Dissociation	Temporary, drastic change in personality, memory, consciousness, or motor behavior to avoid emotional stress. Patient has incomplete or no memory of traumatic event.	A victim of sexual abuse suddenly appears numb and detached when she is exposed to her abuser.

Ego defenses (continued)

IMMATURE DEFENSES	DESCRIPTION	EXAMPLE
Fixation	Partially remaining at a more childish level of development (vs regression).	A surgeon throws a tantrum in the operating room because the last case ran very late.
Idealization	Expressing extremely positive thoughts of self and others while ignoring negative thoughts.	A patient boasts about his physician and his accomplishments while ignoring any flaws.
Identification	Largely unconscious assumption of the characteristics, qualities, or traits of another person or group.	A resident starts putting his stethoscope in his pocket like his favorite attending, instead of wearing it around his neck like before.
Intellectualization	Using facts and logic to emotionally distance oneself from a stressful situation.	In a therapy session, patient diagnosed with cancer focuses only on rates of survival.
Isolation (of affect)	Separating feelings from ideas and events.	Describing murder in graphic detail with no emotional response.
Passive aggression	Demonstrating hostile feelings in a nonconfrontational manner; showing indirect opposition.	Disgruntled employee is repeatedly late to work, but won't admit it is a way to get back at the manager.
Projection	Attributing an unacceptable internal impulse to an external source (vs displacement).	A man who wants to cheat on his wife accuses his wife of being unfaithful.
Rationalization	Proclaiming logical reasons for actions actually performed for other reasons, usually to avoid self-blame.	After getting fired, claiming that the job was not important anyway.
Reaction formation	Replacing a ward-off idea or feeling by an (unconsciously derived) emphasis on its opposite (vs sublimation).	A patient with libidinous thoughts enters a monastery.
Regression	Involuntarily turning back the maturational clock and going back to earlier modes of dealing with the world (vs fixation).	Seen in children under stress such as illness, punishment, or birth of a new sibling (eg, bedwetting in a previously toilet-trained child when hospitalized).
Repression	Involuntarily withholding an idea or feeling from conscious awareness (vs suppression).	A 20-year-old does not remember going to counseling during his parents' divorce 10 years earlier.
Splitting	Believing that people are either all good or all bad at different times due to intolerance of ambiguity. Commonly seen in borderline personality disorder.	A patient says that all the nurses are cold and insensitive but that the doctors are warm and friendly.
MATURE DEFENSES		
Sublimation	Replacing an unacceptable wish with a course of action that is similar to the wish but does not conflict with one's value system (vs reaction formation).	Teenager's aggression toward his father is redirected to perform well in sports.
Altruism	Alleviating negative feelings via unsolicited generosity.	Mafia boss makes large donation to charity.
Suppression	Intentionally withholding an idea or feeling from conscious awareness (vs repression); temporary.	Choosing to not worry about the big game until it is time to play.
Humor	Appreciating the amusing nature of an anxiety-provoking or adverse situation.	Nervous medical student jokes about the boards.
Mature adults wear a SASH .		

► PSYCHIATRY—PATHOLOGY

Infant deprivation effects

Long-term deprivation of affection results in:

- Failure to thrive
- Poor language/socialization skills
- Lack of basic trust
- Reactive attachment disorder (infant withdrawn/unresponsive to comfort)

Deprivation for > 6 months can lead to irreversible changes.

Severe deprivation can result in infant death.

Child abuse

	Physical abuse	Sexual abuse
EVIDENCE	Fractures (eg, ribs, long bone spiral, multiple in different stages of healing), bruises (eg, trunk, ear, neck; in pattern of implement), burns (eg, cigarette, buttocks/thighs), subdural hematomas, retinal hemorrhages. During exam, children often avoid eye contact.	Genital, anal, or oral trauma; STIs; UTIs.
ABUSER	Usually biological mother.	Known to victim, usually male.
EPIDEMIOLOGY	40% of deaths related to child abuse or neglect occur in children < 1 year old.	Peak incidence 9–12 years old.

Child neglect

Failure to provide a child with adequate food, shelter, supervision, education, and/or affection. Most common form of child maltreatment. Evidence: poor hygiene, malnutrition, withdrawal, impaired social/emotional development, failure to thrive. As with child abuse, suspected child neglect must be reported to local child protective services.

Vulnerable child syndrome

Parents perceive the child as especially susceptible to illness or injury. Usually follows a serious illness or life-threatening event. Can result in missed school or overuse of medical services.

Childhood and early-onset disorders

Attention-deficit hyperactivity disorder	Onset before age 12. Limited attention span and poor impulse control. Characterized by hyperactivity, impulsivity, and/or inattention in multiple settings (school, home, places of worship, etc). Normal intelligence, but commonly coexists with difficulties in school. Continues into adulthood in as many as 50% of individuals. Treatment: stimulants (eg, methylphenidate) +/- cognitive behavioral therapy (CBT); alternatives include atomoxetine, guanfacine, clonidine.
Autism spectrum disorder	Characterized by poor social interactions, social communication deficits, repetitive/ritualized behaviors, restricted interests. Must present in early childhood. May be accompanied by intellectual disability; rarely accompanied by unusual abilities (savants). More common in boys. Associated with ↑ head/brain size.
Rett syndrome	X-linked dominant disorder seen almost exclusively in girls (affected males die in utero or shortly after birth). Majority of cases are caused by de novo mutation of <i>MECP2</i> . Symptoms usually become apparent around ages 1–4, including regression characterized by loss of development, loss of verbal abilities, intellectual disability, ataxia, stereotyped hand-wringing. No longer a solitary diagnosis within DSM-5.
Conduct disorder	Repetitive and pervasive behavior violating the basic rights of others or societal norms (eg, aggression to people and animals, destruction of property, theft). After age 18, often reclassified as antisocial personality disorder. Treatment for both: psychotherapy such as CBT.
Oppositional defiant disorder	Enduring pattern of hostile, defiant behavior toward authority figures in the absence of serious violations of social norms. Treatment: psychotherapy such as CBT.
Separation anxiety disorder	Overwhelming fear of separation from home or attachment figure. Can be normal behavior up to age 3–4. May lead to factitious physical complaints to avoid school. Treatment: CBT, play therapy, family therapy.
Tourette syndrome	Onset before age 18. Characterized by sudden, rapid, recurrent, nonrhythmic, stereotyped motor and vocal tics that persist for > 1 year. Coprolalia (involuntary obscene speech) found in only 10–20% of patients. Associated with OCD and ADHD. Treatment: psychoeducation, behavioral therapy. For intractable and distressing tics, high-potency antipsychotics (eg, haloperidol, fluphenazine, pimozide), tetrabenazine, α_2 -agonists (eg, guanfacine, clonidine), or atypical antipsychotics may be used.
Disruptive mood dysregulation disorder	Onset before age 10. Severe and recurrent temper outbursts out of proportion to situation. Child is constantly angry and irritable between outbursts. Treatment: psychostimulants, antipsychotics, CBT.

Orientation

Patient's ability to know who he or she is, where he or she is, and the date and time.

Order of loss: time → place → person.

Common causes of loss of orientation: alcohol, drugs, fluid/electrolyte imbalance, head trauma, hypoglycemia, infection, nutritional deficiencies.

Amnesias**Retrograde amnesia**

Inability to remember things that occurred **before** a CNS insult.

Anterograde amnesia

Inability to remember things that occurred **after** a CNS insult (↓ acquisition of new memory).

Korsakoff syndrome

Amnesia (anterograde > retrograde) caused by vitamin B₁ deficiency and associated destruction of mammillary bodies. Seen in alcoholics as a late neuropsychiatric manifestation of Wernicke encephalopathy. Confabulations are characteristic.

Dissociative amnesia

Inability to recall important personal information, usually subsequent to severe trauma or stress. May be accompanied by **dissociative fugue** (abrupt travel or wandering during a period of dissociative amnesia, associated with traumatic circumstances).

Dissociative disorders**Dissociative identity disorder**

Formerly known as multiple personality disorder. Presence of 2 or more distinct identities or personality states. More common in women. Associated with history of sexual abuse, PTSD, depression, substance abuse, borderline personality, somatoform conditions.

Depersonalization/derealization disorder

Persistent feelings of detachment or estrangement from one's own body, thoughts, perceptions, and actions (depersonalization) or one's environment (derealization).

Delirium

"Waxing and waning" level of consciousness with acute onset; rapid ↓ in attention span and level of arousal. Characterized by disorganized thinking, hallucinations (often visual), illusions, misperceptions, disturbance in sleep-wake cycle, cognitive dysfunction.

Usually 2° to other illness (eg, CNS disease, infection, trauma, substance abuse/withdrawal, metabolic/electrolyte disturbances, hemorrhage, urinary/fecal retention).

Most common presentation of altered mental status in inpatient setting, especially in the intensive care unit and with prolonged hospital stays. Commonly, diffuse slowing EEG.

Treatment is aimed at identifying and addressing underlying condition.

Antipsychotics may be used acutely as needed.

Delirium = changes in **sensorium**.

May be caused by medications (eg, anticholinergics), especially in the elderly.

Reversible.

Dementia

↓ in intellectual function without affecting level of consciousness. Characterized by memory deficits, apraxia, aphasia, agnosia, loss of abstract thought, behavioral/personality changes, impaired judgment. A patient with dementia can develop delirium (eg, patient with Alzheimer disease who develops pneumonia is at ↑ risk for delirium).

Irreversible causes: Alzheimer disease, Lewy body dementia, Huntington disease, Pick disease, cerebral infarct, Wilson disease, Creutzfeldt-Jakob disease, chronic substance abuse (due to neurotoxicity of drugs), HIV.

Reversible causes: hypothyroidism, depression, vitamin deficiency (B₁, B₃, B₁₂), normal pressure hydrocephalus, neurosyphilis.

↑ incidence with age. EEG usually normal.

“**Dementia**” is characterized by **memory** loss. Usually **irreversible**.

In elderly patients, depression and hypothyroidism may present like dementia (pseudodementia). Screen for depression, exclude neurosyphilis with RPR if high clinical suspicion, and measure TSH, B₁₂ levels.

Psychosis

Distorted perception of reality characterized by delusions, hallucinations, and/or disorganized thought/speech. Can occur in patients with medical illness, psychiatric illness, or both.

Delusions

Unique, false, fixed, idiosyncratic beliefs that persist despite the facts and are not typical of a patient's culture or religion (eg, thinking aliens are communicating with you). Types include erotomanic, grandiose, jealous, persecutory, somatic, mixed, and unspecified.

Disorganized thought

Speech may be incoherent (“word salad”), tangential, or derailed (“loose associations”).

Hallucinations

Perceptions in the absence of external stimuli (eg, seeing a light that is not actually present).

Contrast with illusions, misperceptions of real external stimuli. Types include:

- Visual—more commonly a feature of medical illness (eg, drug intoxication) than psychiatric illness.
- Auditory—more commonly a feature of psychiatric illness (eg, schizophrenia) than medical illness.
- Olfactory—often occur as an aura of temporal lobe epilepsy (eg, burning rubber) and in brain tumors.
- Gustatory—rare, but seen in epilepsy.
- Tactile—common in alcohol withdrawal and stimulant use (eg, cocaine, amphetamines), delusional parasitosis, “cocaine crawlies.”
- Hypnagogic—occurs while **going** to sleep. Sometimes seen in narcolepsy.
- Hypnopompic—occurs while waking from sleep (“**pomp**ous upon awakening”). Sometimes seen in narcolepsy.

Schizophrenia

Chronic mental disorder with periods of psychosis, disturbed behavior and thought, and decline in functioning **lasting > 6 months**. Associated with ↑ dopaminergic activity, ↓ dendritic branching.

Diagnosis requires at least 2 of the following, and at least 1 of these should include 1–3 (first 4 are “positive symptoms”):

1. Delusions
2. Hallucinations—often auditory
3. Disorganized speech
4. Disorganized or catatonic behavior
5. Negative symptoms (affective flattening, avolition, anhedonia, asociality, alogia)

Brief psychotic disorder—lasting < 1 month, usually stress related.

Schizophreniform disorder—lasting 1–6 months.

Schizoaffective disorder—Meets criteria for schizophrenia in addition to major mood disorder (major depressive or bipolar). To differentiate from a major mood disorder with psychotic features, patient must have > 2 weeks of hallucinations or delusions without major mood episode.

Frequent cannabis use is associated with psychosis/schizophrenia in teens.

Lifetime prevalence—1.5% (males = females, African Americans = Caucasians). Presents earlier in men (late teens to early 20s vs late 20s to early 30s in women). Patients are at ↑ risk for suicide.

Ventriculomegaly on brain imaging.

Treatment: atypical antipsychotics (eg, risperidone) are first line.

Negative symptoms often persist after treatment, despite resolution of positive symptoms.

Delusional disorder

Fixed, persistent, false belief system **lasting > 1 month**. Functioning otherwise not impaired (eg, a woman who genuinely believes she is married to a celebrity when, in fact, she is not). Can be shared by individuals in close relationships (folie à deux).

Mood disorder

Characterized by an abnormal range of moods or internal emotional states and loss of control over them. Severity of moods causes distress and impairment in social and occupational functioning. Includes major depressive disorder, bipolar disorder, dysthymic disorder, and cyclothymic disorder. Episodic superimposed psychotic features (delusions or hallucinations) may be present.

Manic episode

Distinct period of abnormally and persistently elevated, expansive, or irritable mood and abnormally and persistently ↑ activity or energy **lasting at least 1 week**. Often disturbing to patient.

Diagnosis requires hospitalization or at least 3 of the following (manics **DIG FAST**):

- **D**istractibility
- **I**rresponsibility—seeks pleasure without regard to consequences (hedonistic)
- **G**randiosity—inflated self-esteem
- **F**light of ideas—racing thoughts
- ↑ in goal-directed **A**ctivity/psychomotor **A**gitation
- ↓ need for **S**leep
- **T**alkativeness or pressured speech

Hypomanic episode	Like a manic episode except mood disturbance is not severe enough to cause marked impairment in social and/or occupational functioning or to necessitate hospitalization. No psychotic features. Lasts at least 4 consecutive days.	
Bipolar disorder (manic depression)	Bipolar I defined by presence of at least 1 manic episode +/- a hypomanic or depressive episode. Bipolar II defined by presence of a hypomanic and a depressive episode. Patient's mood and functioning usually return to normal between episodes. Use of antidepressants can precipitate mania. High suicide risk. Treatment: mood stabilizers (eg, lithium, valproic acid, carbamazepine, lamotrigine), atypical antipsychotics. Cyclothymic disorder—milder form of bipolar disorder lasting at least 2 years, fluctuating between mild depressive and hypomanic symptoms.	
Major depressive disorder	Episodes characterized by at least 5 of the 9 diagnostic symptoms lasting ≥ 2 weeks (symptoms must include patient-reported depressed mood or anhedonia). Treatment: CBT and SSRIs are first line. SNRIs, mirtazapine, bupropion can also be considered. Antidepressants are indicated if bipolar disorder is ruled out. Electroconvulsive therapy (ECT) in select patients. Persistent depressive disorder (dysthymia)—depression, often milder, lasting at least 2 years.	Diagnostic symptoms: SIG E CAPS: - Depressed mood - Sleep disturbance - Loss of Interest (anhedonia) - Guilt or feelings of worthlessness - Energy loss and fatigue - Concentration problems - Appetite/weight changes - Psychomotor retardation or agitation - Suicidal ideations Patients with depression typically have the following changes in their sleep stages: ↓ slow-wave sleep ↓ REM latency ↑ REM early in sleep cycle ↑ total REM sleep - Repeated nighttime awakenings - Early-morning awakening (terminal insomnia)
Depression with atypical features	Characterized by mood reactivity (being able to experience improved mood in response to positive events, albeit briefly), “reversed” vegetative symptoms (hypersomnia, hyperphagia), leaden paralysis (heavy feeling in arms and legs), long-standing interpersonal rejection sensitivity. Most common subtype of depression. Treatment: CBT and SSRIs are first line. MAO inhibitors are effective but not first line because of their risk profile.	

Postpartum mood disturbances

Onset within 4 weeks of delivery.

Maternal (postpartum) “blues”

50–85% incidence rate. Characterized by depressed affect, tearfulness, and fatigue starting 2–3 days after delivery. Usually resolves within 10 days. Treatment: supportive. Follow up to assess for possible postpartum depression.

Postpartum depression

10–15% incidence rate. Characterized by depressed affect, anxiety, and poor concentration for ≥ 2 weeks. Treatment: CBT and SSRIs are first line.

Postpartum psychosis

0.1–0.2% incidence rate. Characterized by mood-congruent delusions, hallucinations, and thoughts of harming the baby or self. Risk factors include history of bipolar or psychotic disorder, first pregnancy, family history, recent discontinuation of psychotropic medication. Treatment: hospitalization and initiation of atypical antipsychotic; if insufficient, ECT may be used.

Grief

The five stages of grief per the Kübler-Ross model are denial, anger, bargaining, depression, and acceptance, not necessarily in that order. Other normal grief symptoms include shock, guilt, sadness, anxiety, yearning, and somatic symptoms. Simple hallucinations of the deceased person are common (eg, hearing the deceased speaking). Duration varies widely; usually **< 6 months**. Pathologic grief is persistent and causes functional impairment. Can meet criteria for major depressive episode.

Electroconvulsive therapy

Used mainly for treatment-refractory depression, depression with psychotic symptoms, and acutely suicidality. Produces grand mal seizure in an anesthetized patient. Adverse effects include disorientation, temporary headache, partial anterograde/retrograde amnesia usually resolving in 6 months. Safe in pregnancy.

Risk factors for suicide completion

Sex (male)
Age (young adult or elderly)
Depression
Previous attempt (highest risk factor)
Ethanol or drug use
Rational thinking loss (psychosis)
Sickness (medical illness)
Organized plan
No spouse or other social support
Stated future intent

SAD PERSONS are more likely to complete suicide.

Most common method in US is firearms; access to guns $\uparrow$ risk of suicide completion.

Women try more often; men complete more often.

Anxiety disorder

Inappropriate experience of fear/worry and its physical manifestations (anxiety) incongruent with the magnitude of the perceived stressor. Symptoms interfere with daily functioning and are not attributable to another mental disorder, medical condition, or substance abuse. Includes panic disorder, phobias, generalized anxiety disorder, and selective mutism. Treatment: CBT, SSRIs, SNRIs.

Panic disorder

Defined by recurrent panic attacks (periods of intense fear and discomfort peaking in 10 minutes with at least 4 of the following): **P**alpitations, **P**aresthesias, **d**e**P**ersonalization or derealization, **A**bdominal distress or **N**ausea, **I**ntense fear of dying, **I**ntense fear of losing control or “going crazy,” **L**ight-headedness, **C**hest pain, **C**hills, **C**hoking, **S**weating, **S**haking, **S**hortness of breath. Strong genetic component. ↑ risk of suicide. Treatment: CBT, SSRIs, and venlafaxine are first line. Benzodiazepines occasionally used in acute setting.

PANICS.

Diagnosis requires attack followed by 1 month (or more) of 1 (or more) of the following:

- Persistent concern of additional attacks
- Worrying about consequences of attack
- Behavioral change related to attacks

Symptoms are the systemic manifestations of fear.

Specific phobia

Severe, persistent (≥ 6 months) fear or anxiety due to presence or anticipation of a specific object or situation. Person often recognizes fear is excessive. Can be treated with systematic desensitization.

Social anxiety disorder—exaggerated fear of embarrassment in social situations (eg, public speaking, using public restrooms). Treatment: CBT, SSRIs, venlafaxine. For only occasional anxiety-inducing situations, benzodiazepine or β -blocker.

Agoraphobia—irrational fear/anxiety while facing or anticipating ≥ 2 specific situations (eg, open/closed spaces, lines, crowds, public transport). If severe, patients may refuse to leave their homes. Associated with panic disorder. Treatment: CBT, SSRIs.

Generalized anxiety disorder

Anxiety **lasting** > 6 months unrelated to a specific person, situation, or event. Associated with restlessness, irritability, sleep disturbance, fatigue, muscle tension, difficulty concentrating. Treatment: CBT, SSRIs, SNRIs are first line. Buspirone, TCAs, benzodiazepines are second line.

Adjustment disorder—emotional symptoms (anxiety, depression) that occur within 3 months of an identifiable psychosocial stressor (eg, divorce, illness) **lasting** < 6 months once the stressor has ended. If stressor lasts > 6 months and causes continual impairment, it is GAD. Treatment: CBT, SSRIs.

Obsessive-compulsive disorder

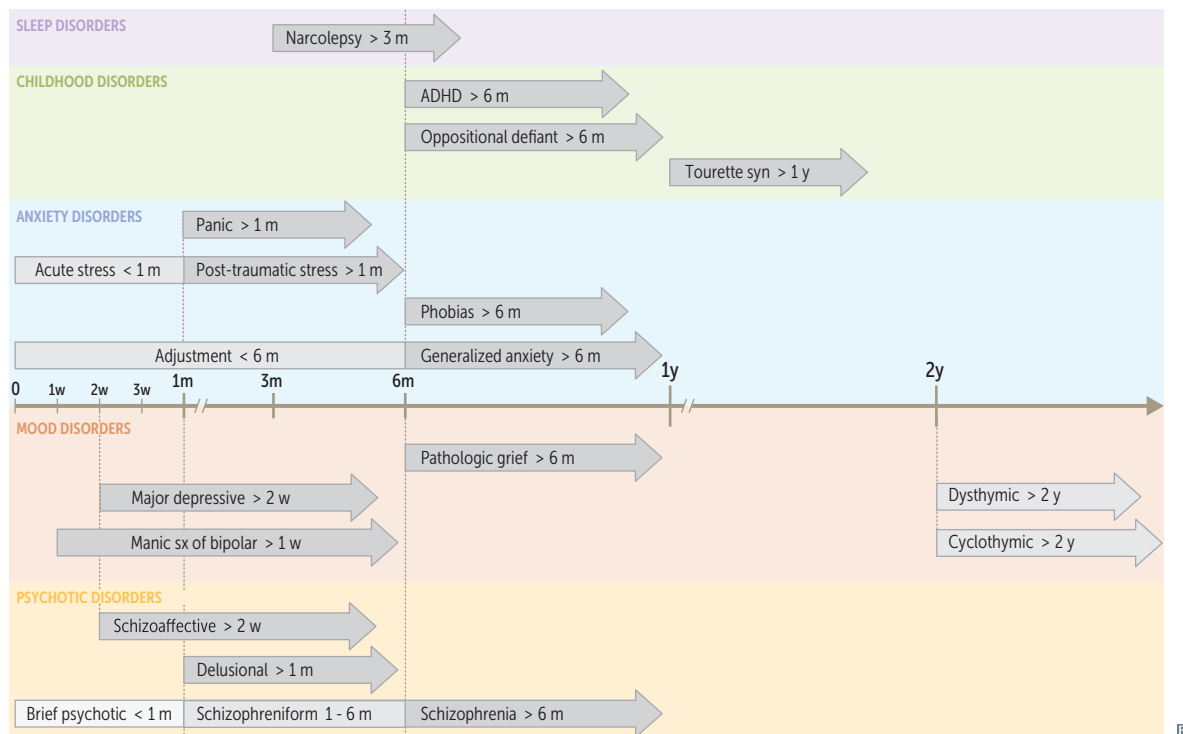
Recurring intrusive thoughts, feelings, or sensations (obsessions) that cause severe distress; relieved in part by the performance of repetitive actions (compulsions). Ego-dystonic: behavior inconsistent with one's own beliefs and attitudes (vs obsessive-compulsive personality disorder). Associated with Tourette syndrome. Treatment: CBT, SSRIs, and clomipramine are first line.

Body dysmorphic disorder—preoccupation with minor or imagined defect in appearance → significant emotional distress or impaired functioning; patients often repeatedly seek cosmetic treatment. Treatment: CBT.

Post-traumatic stress disorder

Exposure to prior trauma (eg, witnessing death, experiencing serious injury or rape) → persistent **H**yperarousal, **A**voidance of associated stimuli, intrusive **R**eexperiencing of the event (nightmares, flashbacks), changes in cognition or mood (fear, horror, **D**istress) (having PTSD is **HARD**).
Disturbance **lasts > 1 month** with significant distress or impaired social-occupational functioning.
Treatment: CBT, SSRIs, and venlafaxine are first line. Prazosin can reduce nightmares.

Acute stress disorder—lasts between 3 days and 1 month. Treatment: CBT; pharmacotherapy is usually not indicated.

Diagnostic criteria by symptom duration**Personality****Personality trait**

An enduring, repetitive pattern of perceiving, relating to, and thinking about the environment and oneself.

Personality disorder

Inflexible, maladaptive, and rigidly pervasive pattern of behavior causing subjective distress and/or impaired functioning; person is usually not aware of problem. Usually presents by early adulthood.

Three clusters, A, B, and C; remember as **W**eird, **W**ild, and **W**orried based on symptoms.

Cluster A personality disorders	Odd or eccentric; inability to develop meaningful social relationships. No psychosis; genetic association with schizophrenia.	“Weird.” Cluster A : A ccusatory, A loof, A wkward.
Paranoid	Pervasive distrust (A ccusatory) and suspiciousness of others and a profoundly cynical view of the world.	
Schizoid	V oluntary social withdrawal (A loof), limited emotional expression, content with social isolation (vs avoidant).	
Schizotypal	Eccentric appearance, odd beliefs or magical thinking, interpersonal A wkwardness.	Schizot y p a l = magical t hinking.
Cluster B personality disorders	Dramatic, emotional, or erratic; genetic association with mood disorders and substance abuse.	“Wild.” Cluster B : B ad, B orderline, flam B oyant, must be the B est
Antisocial	Disregard for and violation of rights of others with lack of remorse, criminality, impulsivity; males > females; must be ≥ 18 years old and have history of conduct disorder before age 15. Conduct disorder if < 18 years old.	Antisocial = sociopath. B ad.
Borderline	Unstable mood and interpersonal relationships, impulsivity, self-mutilation, suicidality, sense of emptiness; females > males; splitting is a major defense mechanism.	Treatment: dialectical behavior therapy. B orderline.
Histrionic	Excessive emotionality and excitability, attention seeking, sexually provocative, overly concerned with appearance.	Flam B oyant.
Narcissistic	Grandiosity, sense of entitlement; lacks empathy and requires excessive admiration; often demands the “best” and reacts to criticism with rage.	Must be the B est.
Cluster C personality disorders	Anxious or fearful; genetic association with anxiety disorders.	“Worried.” Cluster C : C owardly, obsessive- C ompulsive, C lingy.
Avoidant	Hypersensitive to rejection, socially inhibited, timid, feelings of inadequacy, desires relationships with others (vs schizoid).	C owardly.
Obsessive-Compulsive	Preoccupation with order, perfectionism, and control; ego-syntonic: behavior consistent with one’s own beliefs and attitudes (vs OCD).	
Dependent	Submissive and C lingy, excessive need to be taken care of, low self-confidence.	Patients often get stuck in abusive relationships.

Malingering	Patient consciously fakes, profoundly exaggerates, or claims to have a disorder in order to attain a specific 2° (external) gain (eg, avoiding work, obtaining compensation). Poor compliance with treatment or follow-up of diagnostic tests. Complaints cease after gain (vs factitious disorder).
Factitious disorders	Patient consciously creates physical and/or psychological symptoms in order to assume “sick role” and to get medical attention and sympathy (1° [internal] gain).
Factitious disorder imposed on self	Also known as Munchausen syndrome. Chronic factitious disorder with predominantly physical signs and symptoms. Characterized by a history of multiple hospital admissions and willingness to undergo invasive procedures. More common in women and healthcare workers.
Factitious disorder imposed on another	Also known as Munchausen syndrome by proxy. Illness in a child or elderly patient is caused or fabricated by the caregiver. Motivation is to assume a sick role by proxy. Form of child/elder abuse.
Somatic symptom and related disorders	Category of disorders characterized by physical symptoms causing significant distress and impairment. Both illness production and motivation are unconscious drives. Symptoms not intentionally produced or feigned. More common in women.
Somatic symptom disorder	Variety of bodily complaints (eg, pain, fatigue) lasting for months to years. Associated with excessive, persistent thoughts and anxiety about symptoms. May co-occur with medical illness. Treatment: regular office visits with the same physician in combination with psychotherapy.
Conversion disorder (functional neurologic symptom disorder)	Loss of sensory or motor function (eg, paralysis, blindness, mutism), often following an acute stressor; patient is aware of but sometimes indifferent toward symptoms (“la belle indifférence”); more common in females, adolescents, and young adults.
Illness anxiety disorder (hypochondriasis)	Excessive preoccupation with acquiring or having a serious illness, often despite medical evaluation and reassurance; minimal somatic symptoms.

Eating disorders	Most common in young females.
Anorexia nervosa	Excessive dieting, exercise, or binge eating/purging with BMI < 18.5 kg/m ² ; intense fear of gaining weight; and distortion or overvaluation of body image. Associated with ↓ bone density, severe weight loss, metatarsal stress fractures, amenorrhea (due to loss of pulsatile GnRH secretion), lanugo, anemia, electrolyte disturbances. Commonly coexists with depression. Psychotherapy and nutritional rehabilitation are first line; pharmacotherapy includes SSRIs. Refeeding syndrome (↑ insulin → hypophosphatemia → cardiac complications) can occur in significantly malnourished patients.
Bulimia nervosa	Binge eating with recurrent inappropriate compensatory behaviors (eg, self-induced vomiting, using laxatives or diuretics, fasting, excessive exercise) occurring weekly for at least 3 months and overvaluation of body image. Body weight often maintained within normal range. Associated with parotitis, enamel erosion, electrolyte disturbances (eg, hypokalemia, hypochloremia), metabolic alkalosis, dorsal hand calluses from induced vomiting (Russell sign). Treatment: psychotherapy, nutritional rehabilitation, antidepressants.
Binge eating disorder	Regular episodes of excessive, uncontrollable eating without inappropriate compensatory behaviors. ↑ risk of diabetes. Treatment: psychotherapy such as CBT is first-line; SSRIs, lisdexamfetamine.
Gender dysphoria	Persistent cross-gender identification that leads to persistent distress with sex assigned at birth. Transsexualism —desire to live as the opposite sex , often through surgery or hormone treatment. Transvestism —paraphilia, not gender dysphoria. Wearing clothes (eg, vest) of the opposite sex (cross-dressing).
Sexual dysfunction	Includes sexual desire disorders (hypoactive sexual desire or sexual aversion), sexual arousal disorders (erectile dysfunction), orgasmic disorders (anorgasmia, premature ejaculation), sexual pain disorders (dyspareunia, vaginismus). Differential diagnosis includes: ▪ Drug side effects (eg, antihypertensives, antipsychotics, SSRIs, ethanol) ▪ Medical disorders (eg, depression, diabetes, STIs) ▪ Psychological (eg, performance anxiety)

Sleep terror disorder

Inconsolable periods of terror with screaming in the middle of the night; occurs during slow-wave/deep (stage N3) sleep. Most common in children. Occurs during non-REM sleep (no memory of the arousal episode) as opposed to nightmares that occur during REM sleep (memory of a scary dream). Cause unknown, but triggers include emotional stress, fever, or lack of sleep. Usually self limited.

Narcolepsy

Disordered regulation of sleep-wake cycles; 1° characteristic is excessive daytime sleepiness (awaken feeling rested).

Caused by ↓ hypocretin (orexin) production in lateral hypothalamus.

Also associated with:

- Hypnagogic (just before sleep) or hypnopompic (just before awakening) hallucinations.
- Nocturnal and narcoleptic sleep episodes that start with REM sleep (sleep paralysis).
- Cataplexy (loss of all muscle tone following strong emotional stimulus, such as laughter) in some patients.

Strong genetic component. Treatment: daytime stimulants (eg, amphetamines, modafinil) and nighttime sodium oxybate (GHB).

Hypnagogic—going to sleep

Hypnopompic—“pompous upon awakening”

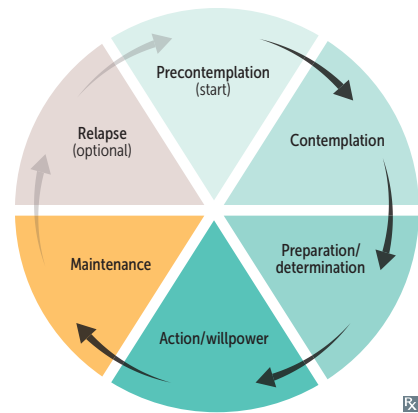
Substance use disorder

Maladaptive pattern of substance use defined as 2 or more of the following signs in 1 year related specifically to substance use:

- Tolerance—need more to achieve same effect
- Withdrawal—manifesting as characteristic signs and symptoms
- Substance taken in larger amounts, or over longer time, than desired
- Persistent desire or unsuccessful attempts to cut down
- Significant energy spent obtaining, using, or recovering from substance
- Important social, occupational, or recreational activities reduced
- Continued use despite knowing substance causes physical and/or psychological problems
- Craving
- Recurrent use in physically dangerous situations
- Failure to fulfill major obligations at work, school, or home
- Social or interpersonal conflicts

Stages of change in overcoming substance addiction

1. **Precontemplation**—not yet acknowledging that there is a problem
2. **Contemplation**—acknowledging that there is a problem, but not yet ready or willing to make a change
3. **Preparation/determination**—getting ready to change behaviors
4. **Action/willpower**—changing behaviors
5. **Maintenance**—maintaining the behavioral changes
6. **Relapse**—returning to old behaviors and abandoning new changes. Does not always happen.



Psychoactive drug intoxication and withdrawal

DRUG	INTOXICATION	WITHDRAWAL
Depressants		
	Nonspecific: mood elevation, ↓ anxiety, sedation, behavioral disinhibition, respiratory depression.	Nonspecific: anxiety, tremor, seizures, insomnia.
Alcohol	Emotional lability, slurred speech, ataxia, coma, blackouts. Serum γ -glutamyltransferase (GGT)—sensitive indicator of alcohol use. AST value is twice ALT value.	Time from last drink: 3–36 hr: minor symptoms similar to other depressants 6–48 hr: withdrawal seizures 12–48 hr: alcoholic hallucinosis (usually visual) 48–96 hr: delirium tremens (DTs) in 5% of cases Treatment: benzodiazepines.
Opioids	Euphoria, respiratory and CNS depression, ↓ gag reflex, pupillary constriction (pinpoint pupils), seizures (overdose). Most common cause of drug overdose death. Treatment: naloxone.	Sweating, dilated pupils, piloerection (“cold turkey”), fever, rhinorrhea, yawning, nausea, stomach cramps, diarrhea (“flu-like” symptoms). Treatment: long-term support, methadone, buprenorphine.
Barbiturates	Low safety margin, marked respiratory depression. Treatment: symptom management (eg, assist respiration, ↑ BP).	Delirium, life-threatening cardiovascular collapse.
Benzodiazepines	Greater safety margin. Ataxia, minor respiratory depression. Treatment: flumazenil (benzodiazepine receptor antagonist, but rarely used as it can precipitate seizures).	Sleep disturbance, depression, rebound anxiety, seizure.
Stimulants		
	Nonspecific: mood elevation, psychomotor agitation, insomnia, cardiac arrhythmias, tachycardia, anxiety.	Nonspecific: post-use “crash,” including depression, lethargy, ↑ appetite, sleep disturbance, vivid nightmares.
Amphetamines	Euphoria, grandiosity, pupillary dilation, prolonged wakefulness and attention, hypertension, tachycardia, anorexia, paranoia, fever. Severe: cardiac arrest, seizures. Treatment: benzodiazepines for agitation and seizures.	
Cocaine	Impaired judgment, pupillary dilation, hallucinations (including tactile), paranoid ideations, angina, sudden cardiac death. Treatment: α -blockers, benzodiazepines. β -blockers not recommended.	
Caffeine	Restlessness, ↑ diuresis, muscle twitching.	Headache, difficulty concentrating, flu-like symptoms.
Nicotine	Restlessness.	Irritability, anxiety, restlessness, difficulty concentrating. Treatment: nicotine patch, gum, or lozenges; bupropion/varenicline.

Psychoactive drug intoxication and withdrawal (continued)

DRUG	INTOXICATION	WITHDRAWAL
Hallucinogens		
Phencyclidine	Violence, impulsivity, psychomotor agitation, nystagmus, tachycardia, hypertension, analgesia, psychosis, delirium, seizures. Trauma is most common complication. Treatment: benzodiazepines, rapid-acting antipsychotic.	
Lysergic acid diethylamide (LSD)	Perceptual distortion (visual, auditory), depersonalization, anxiety, paranoia, psychosis, possible flashbacks.	
Marijuana (cannabinoid)	Euphoria, anxiety, paranoid delusions, perception of slowed time, impaired judgment, social withdrawal, ↑ appetite, dry mouth, conjunctival injection, hallucinations. Pharmaceutical form is dronabinol : used as antiemetic (chemotherapy) and appetite stimulant (in AIDS).	Irritability, anxiety, depression, insomnia, restlessness, ↓ appetite.
MDMA (ecstasy)	Hallucinogenic stimulant: euphoria, disinhibition, hyperactivity, distorted sensory and time perception, teeth clenching. Life-threatening effects include hypertension, tachycardia, hyperthermia, hyponatremia, serotonin syndrome.	Depression, fatigue, change in appetite, difficulty concentrating, anxiety.
Heroin detoxification medications		
Methadone	Users at ↑ risk for hepatitis, HIV, abscesses, bacteremia, right-heart endocarditis.	
Buprenorphine + naloxone	Long-acting oral opiate used for heroin detoxification or long-term maintenance therapy.	
Naltrexone	Sublingually, buprenorphine (partial agonist) is absorbed and used for maintenance therapy. Naloxone (antagonist, not orally bioavailable) is added to lower IV abuse potential.	
Alcoholism	Long-acting opioid antagonist used for relapse prevention once detoxified.	
Alcoholism	Physiologic tolerance and dependence on alcohol with symptoms of withdrawal when intake is interrupted. Complications: alcoholic cirrhosis, hepatitis, pancreatitis, peripheral neuropathy, testicular atrophy. Treatment: disulfiram (to condition the patient to abstain from alcohol use), acamprosate, naltrexone, supportive care. Support groups such as Alcoholics Anonymous are helpful in sustaining abstinence and supporting patient and family.	
Wernicke-Korsakoff syndrome	Caused by vitamin B ₁ deficiency. Triad of confusion, ophthalmoplegia, ataxia (Wernicke encephalopathy). May progress to irreversible memory loss, confabulation, personality change (Korsakoff syndrome). Symptoms may be precipitated by giving dextrose before administering vitamin B ₁ to a patient with thiamine deficiency. Associated with periventricular hemorrhage/necrosis of mammillary bodies. Treatment: IV vitamin B ₁ .	

Delirium tremens

Life-threatening alcohol withdrawal syndrome that peaks 2–4 days after last drink.

Characterized by autonomic hyperactivity (eg, tachycardia, tremors, anxiety, seizures), electrolyte disturbances, respiratory alkalosis. Classically occurs in hospital setting (eg, 2–4 days postsurgery) in alcoholics not able to drink as inpatients. Treatment: benzodiazepines (eg, chlordiazepoxide, lorazepam, diazepam).

► PSYCHIATRY—PHARMACOLOGY

Preferred medications for selected psychiatric conditions

PSYCHIATRIC CONDITION	PREFERRED DRUGS
ADHD	Stimulants (methylphenidate, amphetamines)
Alcohol withdrawal	Benzodiazepines (eg, chlordiazepoxide, lorazepam, diazepam)
Bipolar disorder	Lithium, valproic acid, carbamazepine, lamotrigine, atypical antipsychotics
Bulimia nervosa	SSRIs
Depression	SSRIs
Generalized anxiety disorder	SSRIs, SNRIs
Obsessive-compulsive disorder	SSRIs, venlafaxine, clomipramine
Panic disorder	SSRIs, venlafaxine, benzodiazepines
PTSD	SSRIs, venlafaxine
Schizophrenia	Atypical antipsychotics
Social anxiety disorder	SSRIs, venlafaxine
Tourette syndrome	Performance only: β -blockers, benzodiazepines Antipsychotics (eg, fluphenazine, pimozide), tetrabenazine

Central nervous system stimulants

Methylphenidate, dextroamphetamine, methamphetamine.

MECHANISM

↑ catecholamines in the synaptic cleft, especially norepinephrine and dopamine.

CLINICAL USE

ADHD, narcolepsy, appetite control.

ADVERSE EFFECTS

Nervousness, agitation, anxiety, insomnia, anorexia, tachycardia, hypertension.

Typical antipsychotics	Haloperidol, pimozide, trifluoperazine, fluphenazine, thioridazine, chlorpromazine.	
MECHANISM	Block dopamine D ₂ receptor (↑ cAMP).	
CLINICAL USE	Schizophrenia (1° positive symptoms), psychosis, bipolar disorder, delirium, Tourette syndrome, Huntington disease, OCD.	
POTENCY	High potency: Trifluoperazine, Fluphenazine, Haloperidol (Try to Fly High)—neurologic side effects (eg, extrapyramidal symptoms [EPS]). Low potency: Chlorpromazine, Thioridazine (Cheating Thieves are low)—anticholinergic, antihistamine, α₁-blockade effects.	
ADVERSE EFFECTS	Lipid soluble → stored in body fat → slow to be removed from body. Endocrine: dopamine receptor antagonism → hyperprolactinemia → galactorrhea, oligomenorrhea, gynecomastia. Metabolic: dyslipidemia, weight gain, hyperglycemia. Antimuscarinic: dry mouth, constipation. Antihistamine: sedation. α₁-blockade: orthostatic hypotension. Cardiac: QT prolongation. Ophthalmologic: Chlorpromazine—Corneal deposits; Thioridazine—reTinal deposits. EPS—ADAPT: - Hours to days: Acute Dystonia (muscle spasm, stiffness, oculogyric crisis). - Days to months: Akathisia (restlessness), Parkinsonism (bradykinesia). - Months to years: Tardive dyskinesia (orofacial chorea). Treatment: benztropine (acute dystonia, tardive dyskinesia), benzodiazepines, β-blockers (akathisia). Neuroleptic malignant syndrome (NMS)—Malignant FEVER: Myoglobinuria, Fever, Encephalopathy, unstable Vitals, ↑ Enzymes, muscle Rigidity. Treatment: dantrolene, D₂ agonist (eg, bromocriptine).	
Atypical antipsychotics	Aripiprazole, asenapine, clozapine, olanzapine, quetiapine, iloperidone, paliperidone, risperidone, lurasidone, ziprasidone.	
MECHANISM	Not completely understood. Most are D ₂ antagonists; aripiprazole is D ₂ partial agonist. Varied effects on 5-HT ₂ , dopamine, and α- and H ₁ -receptors.	
CLINICAL USE	Schizophrenia—both positive and negative symptoms. Also used for bipolar disorder, OCD, anxiety disorder, depression, mania, Tourette syndrome.	Use clozapine for treatment-resistant schizophrenia or schizoaffective disorder and for suicidality in schizophrenia.
ADVERSE EFFECTS	All—prolonged QT interval, fewer EPS and anticholinergic side effects than typical antipsychotics. “-pines”—metabolic syndrome (weight gain, diabetes, hyperlipidemia). Clozapine—agranulocytosis (monitor WBCs frequently) and seizures (dose related). Risperidone—hyperprolactinemia (amenorrhea, galactorrhea, gynecomastia).	
		Olanzapine → Obesity Must watch bone marrow clozely with clozapine.

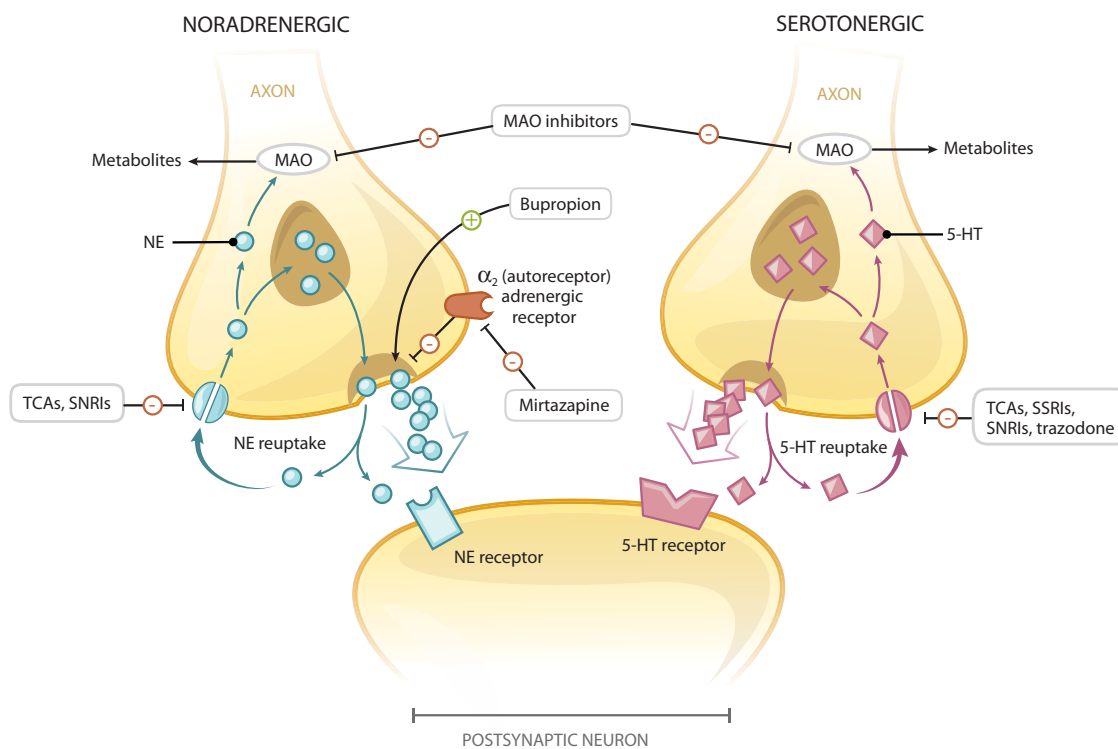
Lithium

MECHANISM	Not established; possibly related to inhibition of phosphoinositol cascade.	LiTHIUM: Low T hyroid (hypothyroidism) H earth (Ebstein anomaly) I nsipidus (nephrogenic diabetes insipidus) U nwanted M ovements (tremor)
CLINICAL USE	Mood stabilizer for bipolar disorder; blocks relapse and acute manic events.	
ADVERSE EFFECTS	Tremor, hypothyroidism, polyuria (causes nephrogenic diabetes insipidus), teratogenesis. Causes Ebstein anomaly in newborn if taken by pregnant mother. Narrow therapeutic window requires close monitoring of serum levels. Almost exclusively excreted by kidneys; most is reabsorbed at PCT with Na ⁺ . Thiazides (and other nephrotoxic agents) are implicated in lithium toxicity.	

Buspirone

MECHANISM	Stimulates 5-HT _{1A} receptors.	I'm always anxious if the bus will be on time, so I take buspirone .
CLINICAL USE	Generalized anxiety disorder. Does not cause sedation, addiction, or tolerance. Takes 1–2 weeks to take effect. Does not interact with alcohol (vs barbiturates, benzodiazepines).	

Antidepressants



Selective serotonin reuptake inhibitors

Fluoxetine, fluvoxamine, paroxetine, sertraline, escitalopram, citalopram.

MECHANISM	Inhibit 5-HT reuptake.	It normally takes 4–8 weeks for antidepressants to have an effect.
CLINICAL USE	Depression, generalized anxiety disorder, panic disorder, OCD, bulimia, social anxiety disorder, PTSD, premature ejaculation, premenstrual dysphoric disorder.	
ADVERSE EFFECTS	Fewer than TCAs. GI distress, SIADH, sexual dysfunction (anorgasmia, ↓ libido).	

Serotonin-norepinephrine reuptake inhibitors

Venlafaxine, desvenlafaxine, duloxetine, levomilnacipran, milnacipran.

MECHANISM	Inhibit 5-HT and norepinephrine reuptake.
CLINICAL USE	Depression, general anxiety disorder, diabetic neuropathy. Venlafaxine is also indicated for social anxiety disorder, panic disorder, PTSD, OCD. Duloxetine is also indicated for fibromyalgia.
ADVERSE EFFECTS	↑ BP most common; also stimulant effects, sedation, nausea.

Serotonin syndrome

Can occur with any drug that ↑ 5-HT (eg, MAOIs, SSRIs, SNRIs, TCAs, tramadol, ondansetron, triptans, linezolid, MDMA, dextromethorphan). Characterized by **3 A's**: neuromuscular hyperActivity (clonus, hyperreflexia, hypertonia, tremor, seizure), Autonomic stimulation (hyperthermia, diaphoresis, diarrhea), and Agitation. Treatment: cyproheptadine (5-HT₂ receptor antagonist).

Tricyclic antidepressants

Amitriptyline, nortriptyline, imipramine, desipramine, clomipramine, doxepin, amoxapine.

MECHANISM	Inhibit NE and 5-HT reuptake.
CLINICAL USE	Major depression, OCD (clomipramine), peripheral neuropathy, chronic pain, migraine prophylaxis. Nocturnal enuresis (imipramine, although adverse effects may limit use).
ADVERSE EFFECTS	Sedation, α ₁ -blocking effects including postural hypotension, and atropine-like (anticholinergic) side effects (tachycardia, urinary retention, dry mouth). 3° TCAs (amitriptyline) have more anticholinergic effects than 2° TCAs (nortriptyline). Can prolong QT interval. Tri-C's : Convulsions, Coma, Cardiotoxicity (arrhythmia due to Na ⁺ channel inhibition); also respiratory depression, hyperpyrexia. Confusion and hallucinations in elderly due to anticholinergic side effects (nortriptyline better tolerated in the elderly). Treatment: NaHCO ₃ to prevent arrhythmia.

Monoamine oxidase inhibitors

Tranylcypromine, Phenelzine, Isocarboxazid, Selegiline (selective MAO-B inhibitor).
(MAO Takes Pride In Shanghai).

MECHANISM	Nonselective MAO inhibition ↑ levels of amine neurotransmitters (norepinephrine, 5-HT, dopamine).
CLINICAL USE	Atypical depression, anxiety. Parkinson disease (selegiline).
ADVERSE EFFECTS	CNS stimulation; hypertensive crisis, most notably with ingestion of tyramine, which is found in many foods such as aged cheese and wine. Tyramine displaces other neurotransmitters (eg, NE) into the synaptic cleft → ↑ sympathetic stimulation. Contraindicated with SSRIs, TCAs, St. John's wort, meperidine, dextromethorphan (to prevent serotonin syndrome). Wait 2 weeks after stopping MAO inhibitors before starting serotonergic drugs or stopping dietary restrictions.

Atypical antidepressants

Bupropion	Inhibits reuptake of NE and dopamine. Also used for smoking cessation. Toxicity: stimulant effects (tachycardia, insomnia), headache, seizures in anorexic/bulimic patients. May help alleviate sexual dysfunction.
Mirtazapine	α_2 -antagonist (↑ release of NE and 5-HT), potent 5-HT ₂ and 5-HT ₃ receptor antagonist and H ₁ antagonist. Toxicity: sedation (which may be desirable in depressed patients with insomnia), ↑ appetite, weight gain (which may be desirable in elderly or anorexic patients), dry mouth.
Trazodone	Primarily blocks 5-HT ₂ , α_1 -adrenergic, and H ₁ receptors; also weakly inhibits 5-HT reuptake. Used primarily for insomnia, as high doses are needed for antidepressant effects. Toxicity: sedation, nausea, priapism, postural hypotension. Called traZZZobone due to sedative and male-specific side effects.
Varenicline	Nicotinic ACh receptor partial agonist. Used for smoking cessation. Toxicity: sleep disturbance, may depress mood.
Vilazodone	Inhibits 5-HT reuptake; 5-HT _{1A} receptor partial agonist. Used for major depressive disorder and generalized anxiety disorder (off-label). Toxicity: headache, diarrhea, nausea, ↑ weight, anticholinergic effects. May cause serotonin syndrome if taken with other serotonergic agents.
Vortioxetine	Inhibits 5-HT reuptake; 5-HT _{1A} receptor agonist and 5-HT ₃ receptor antagonist. Used for major depressive disorder. Toxicity: nausea, sexual dysfunction, sleep disturbances (abnormal dreams), anticholinergic effects. May cause serotonin syndrome if taken with other serotonergic agents.